



# Near-Complete Proviral Genome Sequence of Reticuloendotheliosis Virus Isolated from an Attwater's Prairie Chicken (*Tympanuchus cupido attwateri*)

Bianca Willis,<sup>a,d</sup> Brittany Stewart,<sup>a</sup> Faith Cox,<sup>a</sup> Tiffany Lujan,<sup>a,d</sup> Holly Haefele,<sup>b</sup> Camila Romano,<sup>c</sup> Jeff Brady,<sup>d</sup>  Dustin Edwards<sup>a</sup>

<sup>a</sup>Department of Biological Sciences, Tarleton State University, Stephenville, Texas, USA

<sup>b</sup>Fossil Rim Wildlife Center, Glen Rose, Texas, USA

<sup>c</sup>Instituto de Medicina Tropical de São Paulo e Hospital das Clínicas da Faculdade de Medicina, Universidade de São Paulo, São Paulo, Brazil

<sup>d</sup>Texas A&M AgriLife Research, Stephenville, Texas, USA

**ABSTRACT** We report the near-complete proviral genome sequence of a reticuloendotheliosis virus isolated and propagated from an endangered Attwater's prairie chicken (*Tympanuchus cupido attwateri*) during a 2016–2017 outbreak at a captive breeding facility.

Reticuloendotheliosis virus (REV), a member of the *Retroviridae* family and *Gammaretrovirus* genus, is an immunosuppressive avian virus that infects B lymphocytes and has been detected in Anseriformes, Galliformes, and Passeriformes (1). In 2016 to 2017, the virus was the leading cause of death among adult endangered Attwater's prairie chickens (APCs) (*Tympanuchus cupido attwateri*) at the Fossil Rim Wildlife Center (FRWC) captive breeding facility in Glen Rose, Texas (2). REV was previously isolated and sequenced from APC-566 (GenBank accession number [DQ387450](#)) during an outbreak at the facility 10 years earlier (3). We performed next-generation sequencing of a 2017 isolate for comparison between outbreaks. Blood from APC-2794, diagnosed as REV positive by quantitative PCR (qPCR) at the Texas A&M Veterinary Medical Diagnostic Laboratory (College Station, TX), was drawn into EDTA-coated tubes by FRWC veterinarians in January 2017 as part of normal practices. Blood was received and diluted 1:3 in phosphate-buffered saline (PBS), and mononuclear cells were isolated by Ficoll-Paque density gradient centrifugation. The buffy coat layer was aspirated, washed in PBS, and cocultivated for 5 days with chicken embryonic fibroblast DF-1 cells (CRL-12203; ATCC) with 5% CO<sub>2</sub> at 39°C in Dulbecco's modified Eagle medium (11965-092; Gibco) supplemented with 10% fetal bovine serum (20082-147; Gibco) (4). After incubation, supernatant containing released REV was collected, passed through 0.22-μm filters, and placed on fresh DF-1 cells for 5 days, which was repeated for a total of three passages. After the final passage, cells were washed with PBS, pelleted by centrifugation, resuspended in cell culture freezing medium (12646-010; Gibco), aliquoted, and stored in liquid nitrogen. Supernatant containing REV was stored at –80°C, and a frozen aliquot was submitted to Charles River Avian Services (Wilmington, MA) and determined to contain a 50% tissue culture infective dose (TCID<sub>50</sub>) of  $6.8 \times 10^5$  virions/mL.

Genomic DNA containing provirus was extracted by incubating approximately  $1 \times 10^6$  REV-infected DF-1 cells with 75 μL HotSHOT alkaline lysis reagent at 95°C for 15 min and was cooled on ice for 5 min before the addition of 75 μL of HotSHOT neutralization solution (5). A genomic sequencing library was prepared at Texas A&M AgriLife Research (Stephenville, TX) using KAPA HiFi HotStart ReadyMix (07958927001; Roche) and primers with Illumina adapters to amplify three overlapping sections, i.e., nucleotide positions 1 to 5075, 1500 to 7295, and 7052 to 8295 (3). Resolved PCR products were excised, pooled, tagged using the Nextera XT DNA library preparation kit (FC-131-1096; Illumina), and sequenced with an Illumina MiSeq instrument at Texas A&M Genomics and Bioinformatics Services (College Station, TX). Sequencing produced a total of 35,372 paired-end reads of 300-base read

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Address correspondence to Dustin Edwards, [dc Edwards@tarleton.edu](mailto:dc Edwards@tarleton.edu).

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**TABLE 1** Reference REV isolates used for strain comparisons

| GenBank accession no. | Host    | Country  | Year of isolation | Identity (%) <sup>a</sup> |
|-----------------------|---------|----------|-------------------|---------------------------|
| OL857288              | APC     | USA      | 2017              |                           |
| KJ756349              | Turkey  | USA      | 2014              | 100.00                    |
| DQ387450              | APC     | USA      | 2005              | 99.99                     |
| MF631845              | Chicken | Thailand | 2013              | 99.93                     |
| GQ415646              | Chicken | China    | 2009              | 99.89                     |
| FJ439119              | Goose   | Taiwan   | 2006              | 99.89                     |
| JQ804915              | Duck    | China    | 2011              | 99.84                     |

<sup>a</sup> Percent identity is based on nucleotide alignments obtained from BLASTn.

length. Primer sequences and low-quality and short (<30-bp) reads were trimmed. A consensus sequence was assembled from sequence reads of two libraries for high coverage, prepared from the same DNA extraction, and a reference (GenBank accession number [DQ387450](#)) using the CLC Genomics Workbench v6 assembly to reference tool (mismatch cost=1, indels cost=2, minimum fraction of similarity between reads and reference=90%). A total of 92.4% of the reads were mapped (average coverage, 1,948×). Genome annotation was formatted with SeqBuilder Pro v17.2.1. All tools were run with default parameters unless indicated otherwise. The assembled sequence is 8,246 bp, representing 99.5% of the genome, and has a G+C content of 52.3%. Nucleotide alignment of entire genome sequences with BLASTn (<https://blast.ncbi.nlm.nih.gov>) (6) showed that the virus is well conserved, with 99.9% nucleotide identity to strain APC-566 with a single nucleotide polymorphism at position 6653 (Table 1). APC-2794 shared 100% nucleotide identity with strain 104865 (GenBank accession number [KJ756349](#)), which was isolated from a domestic backyard turkey.

**Data availability.** This sequence was deposited under GenBank accession number [OL857288](#). Raw reads are available under SRA accession numbers [SRX11820216](#) and [SRX13461851](#).

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